

CSCE 650 VIRTUAL REALITY · MODULE 09 · LECTURE NOTES

Interaction and Input

Hands in a world that does not push back.

MODULE OUTCOMES

- Compare direct manipulation with ray-based selection.
- Apply Fitts's law in three dimensions.
- Design affordances a first-time user discovers unaided.
- Explain why hands pass through objects and what to do about it.
- Compare hand tracking with controllers honestly.

1 Two interaction models

	Direct manipulation	Ray / pointer
Method	Reach out and touch the object with a virtual hand.	Project a ray from the hand or controller and select what it hits.
Discoverability	Immediate. No instruction required; people reach for things.	Requires a moment of learning, after which it feels natural.
Range	Arm's length.	Unlimited.
Precision	Good at close range; limited by hand steadiness.	Falls with distance — a fixed angular error in hand orientation becomes a larger positional error further away.
Fatigue	High if sustained — the 'gorilla arm' problem. Holding an arm out is tiring within a couple of minutes.	Lower — the hand can rest at the side or on a lap.
Best for	Close, physical, tactile interaction — picking things up, operating controls.	Menus, distant objects, and seated use.

Most applications need both, and switching between them cleanly — so the user is never unsure which model is active — is a genuine design problem. A common solution is proximity: direct manipulation when the hand is near an object, ray pointing otherwise, with a visible transition.

FITTS'S LAW APPLIES, WITH ANGULAR SIZE AS THE MEASURE

Time to acquire a target grows with distance and falls with target size — the same relationship that governs desktop interfaces, and it holds in three dimensions.

For ray pointing, angular size is what matters. A large object far away subtends a small angle and is therefore a small target, however large it looks in world units. This is why distant menus are frustrating even when they appear big.

And an unsupported hand is substantially less precise than a mouse. There is no surface to rest on, the arm is a long lever amplifying shoulder tremor, and none of it is filtered by friction the way a mouse on a desk is.

So: make targets angularly large, bring interactive content within reach where possible, and filter the input. Interface elements sized reasonably for a monitor are frequently far too small in VR, and this is one of the most common failures in ported applications.

- **Filter the pose.** A one-euro filter, or any speed-adaptive low-pass, removes hand tremor while adding negligible lag during deliberate motion. This is cheap and noticeably improves precision.
- **Snapping and magnetism.** Pull the ray toward valid targets, or snap a held object into a socket when it is close and roughly aligned. Users read this as the system being helpful rather than as imprecision being hidden.
- **Make hit volumes larger than visual volumes.** The user judges by what they see, so a generous collider costs nothing perceptually and measurably improves acquisition.
- **Design for a supported arm where you can** — elbows on a desk, or interactions brief enough that sustained reaching is not required. **Gorilla arm is a real constraint**, not a theoretical one.
- **Avoid tasks needing sub-centimetre precision** unless the user has a physical surface to brace against. Threading a virtual needle in mid-air is not a reasonable thing to ask.

2 Grabbing

YOUR HAND WILL PASS THROUGH THE OBJECT

There is nothing to stop it. The virtual object is solid; the physical hand is moving through empty air and is constrained by nothing.

If the virtual hand tracks the real hand exactly, it penetrates geometry — which Module 01 identified as the single most common break of presence, and the one users test for deliberately within the first minute.

If the virtual hand stops at the surface, it diverges from the real hand's position. The user's proprioception says the hand is at one place and the display says another.

The divergence is the lesser evil, and it is the standard solution: a physics-driven virtual hand that collides with the world and stops at surfaces, with the true tracked position either hidden entirely or shown as a faint ghost. **Users accept this readily** — far more readily than they accept a hand inside a table — and most do not consciously notice it.

Approach	Behaviour	Appropriate for
Snap to a fixed pose	The object jumps to a predetermined position and orientation in the hand.	Tools and weapons , where a predictable grip matters more than a natural one. A pistol should always be held the same way.
Preserve the offset	The object maintains its pose relative to the hand at the moment of grabbing.	Natural for most objects . Picking up a mug by its rim keeps it held by the rim.
Physics joint	A spring or configurable joint connects the object to the hand, so the object is driven toward the hand rather than attached to it.	Heavy objects, and anything that should collide with the world . The object can be blocked by a wall while the hand continues, which is what makes a held object feel like it has mass. More work than it appears , and it is what distinguishes a convincing object from a floating one.
Two-handed	Both hands constrain the object — one gives position, the pair give orientation.	Large objects, rifles, steering. Noticeably more precise for aiming than one hand.

3 Affordances

NOBODY READS INSTRUCTIONS IN VR

Text is uncomfortable to read in a headset (Module 02), the user is occupied looking around a new environment, and there is no manual to consult. Any instruction you write will be skipped.

So interactions must be discoverable from the objects themselves. A grabbable object must look grabbable; a handle must read as a handle; a button must look pressable.

Highlight on proximity is the simplest and most effective technique available: objects that visibly respond when a hand approaches teach their own interactivity, immediately and without words. If you implement one affordance technique, implement this one.

Use real-world forms. A lever, a dial, a toggle switch, a doorknob, a drawer handle — people already know what to do with these, and that knowledge transfers completely and instantly into VR. **Inventing novel interaction metaphors is expensive** and almost never repays the cost.

- **Visual feedback:** highlight on hover, a visible change on grab, an animation on release. This is the minimum, and alone it is thin.
- **Audio feedback:** a click on contact, a sound on grab, a thud on release. **Cheap and disproportionately effective** — a click when a virtual hand touches a virtual surface does a remarkable amount of work toward making the surface feel present.
- **Haptic feedback:** a short controller pulse on contact. The actuator is crude and cannot simulate texture or resistance, and it still substantially improves the sense of having touched something.
- **And the three compound.** Visual plus audio plus haptic together feel considerably more solid than the sum of their individual contributions — the brain is integrating across modalities, and agreement across modalities is what it treats as evidence of reality.
- **Because there is no force feedback,** these three are the entire vocabulary available for signalling contact. A hand that touches a wall and nothing happens is a hand that has touched nothing.

4 Hand tracking and controllers

	Controllers	Hand tracking
Precision	High and consistent.	Lower, and it varies with hand pose and lighting.
Discrete input	Buttons, triggers, grips — unambiguous and plentiful.	None. Only gestures, which must be classified. See below.
Haptics	Yes.	None at all.
Occlusion	Tracked through occlusion by the controller's own IMU for a useful period.	Fails when the hand is hidden — behind the body, behind the other hand, outside the camera field.

	Controllers	Hand tracking
Learning	A moment to learn which button does what.	Immediate. People already have hands.
Fatigue	Lower — there is something to hold and rest.	Higher — nothing to grip, and hands must stay in the tracking volume.
Feels natural	Less so.	Considerably more, when it works.

Hand tracking demonstrates better and controllers work better, which is an uncomfortable conclusion and a reliable one. A five-minute demonstration with hand tracking is magical; a two-hour application is markedly easier with controllers.

THE MISSING BUTTON PROBLEM

A controller provides an unambiguous binary input. The trigger is pressed or it is not, the system knows which, and the user can feel which.

A hand provides nothing of the kind. Pinch detection is a classifier operating on an estimated hand pose with a threshold, and like every classifier it has false positives and false negatives. Fingers that happen to come close produce a grab; a deliberate pinch at an awkward angle does not.

So every interaction must tolerate misdetection: accidental grabs, failed grabs, and releases that the user intended but the system did not register. Designs that assume reliable discrete input will fail in ways that feel like the user's fault but are not.

And there is no haptic confirmation. The user cannot feel whether their action registered, so visual and audio feedback must carry the entire load — which makes §3's feedback discipline not merely desirable but structurally necessary.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
LaValle — Virtual Reality, Chapter 10.2 (free)	Interaction mechanics and the research framing of manipulation in VR.
Meta — interaction and hand tracking design guidelines (free)	The practical rules, including hit volumes, gesture reliability, and the feedback recommendations of §3.
Casiez et al. — 1€ Filter (free, with code)	The input filter of §1. A page of code, immediately useful, and it works well.
Bowman et al. — 3D User Interfaces: Theory and Practice	The standard text on 3D interaction. Library, not free, and the selection and manipulation taxonomy is worth knowing.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Implement both direct manipulation and ray pointing, with a clean transition between them based on proximity. Test whether users ever become confused about which is active.
- 2 Measure Fitts's law in VR: time selections of targets at several angular sizes and distances, and fit the model.
- 3 Implement a one-euro filter on controller pose and measure the improvement in a precision task.
- 4 Implement a hand that tracks exactly and one that stops at surfaces. Have six people use both and ask which felt wrong.
- 5 Implement all four grab approaches from §2 and compare them on the same object.
- 6 Implement a physics-joint grab and demonstrate a held object colliding with a wall.
- 7 Build an interaction with no instructions at all and watch a novice attempt it. Record every moment of confusion.
- 8 Add visual, then audio, then haptic feedback to a single interaction, and have people rate solidity after each addition.
- 9 Compare hand tracking and controllers on the same task, measuring completion time and error rate.
- 10 Measure the false positive and false negative rate of pinch detection across several hand orientations.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Compare direct manipulation and ray pointing on six axes.
- 2 How does Fitts's law apply in VR, and what is the relevant measure of target size?
- 3 Give five techniques for helping an unsupported hand.
- 4 Why does the virtual hand penetrate objects, and what is the standard solution?
- 5 Name four grab implementations and say what each suits.
- 6 Why does nobody read instructions in VR, and what replaces them?
- 7 Name the three feedback channels and say why they compound.
- 8 Compare hand tracking and controllers on seven axes.
- 9 What is the missing button problem, and what two consequences follow?